ITM Skills University
School of Information Technology

PROJECT REPORT

Case Study No. 50 | HTML5, CSS3 & JavaScript

DevDocs: Software Architecture Patterns

A Web-Based Subscription Portal for Professional Software Architects

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1. General Description

DevDocs is a web-based subscription portal designed for professional software architects and senior engineers. The application serves as a centralised knowledge hub that provides access to verified system design blueprints, architecture patterns, and real-time performance metrics. Built entirely using front-end technologies — HTML5, CSS3, and Vanilla JavaScript — the portal simulates a production-grade SaaS product with a clean, developer-centric interface.

The platform addresses the growing demand among software professionals who need structured, reliable documentation on how large-scale systems are designed, the trade-offs involved, and the infrastructure costs associated with each architectural decision. DevDocs bridges the gap between theoretical computer science and real-world system design practice.

The application features a fully responsive dark/light mode interface, interactive navigation, real-time search and filtering capabilities, and a tiered subscription model — all implemented without any back-end framework or external library, demonstrating strong command of native web technologies.

2. Project Overview

Project Title	DevDocs — Software Architecture Patterns Portal
Case Study No.	50
Industry	Technical Training / SaaS / EdTech
Technologies	HTML5, CSS3, JavaScript (Vanilla)
Total Blueprints	16 Architecture Patterns
Performance Metrics	15 Real-time KPI Cards
Subscription Plans	3 Tiers: Starter, Pro, Enterprise
Theme Support	Dark Mode & Light Mode with localStorage persistence

3. Problem Statement

Software architects and senior engineers frequently encounter a fragmented landscape of documentation. Architecture patterns are spread across books, blog posts, and informal sources — often lacking cost estimates, trade-off analyses, and technology stack guidance in a single unified reference.

DevDocs addresses this by providing a subscription-gated portal where verified system design blueprints are presented in a standardised card format. Each blueprint includes the recommended technology stack, clear pros and cons, expected user scale, latency classification, and estimated monthly infrastructure cost — enabling professionals to make informed architectural decisions quickly and confidently.

4. Design Approach & Layout Structure

4.1 Overall Layout

The application follows a classic Sidebar + Main Content layout, typical of professional SaaS dashboards. The layout is structured as follows:

- Sidebar (280px fixed): Contains the DevDocs logo, three navigation links (Patterns, Pricing, Performance), and a user profile button pinned to the bottom.
- Main Content Area: Dynamically switches between three data sections using JavaScript-driven visibility toggling.
- Landing Page: A full-screen overlay presented on first load that fades out and is removed from the DOM on portal entry.
- Profile Panel: A 400px slide-in right panel for user account details, activity statistics, and theme preferences.

4.2 Responsive Design

The layout adapts for viewports narrower than 1100px by auto-collapsing the sidebar. A fixed MENU button is always accessible to toggle the sidebar. The pricing grid switches from a three-column layout to single-column on screens below 900px.

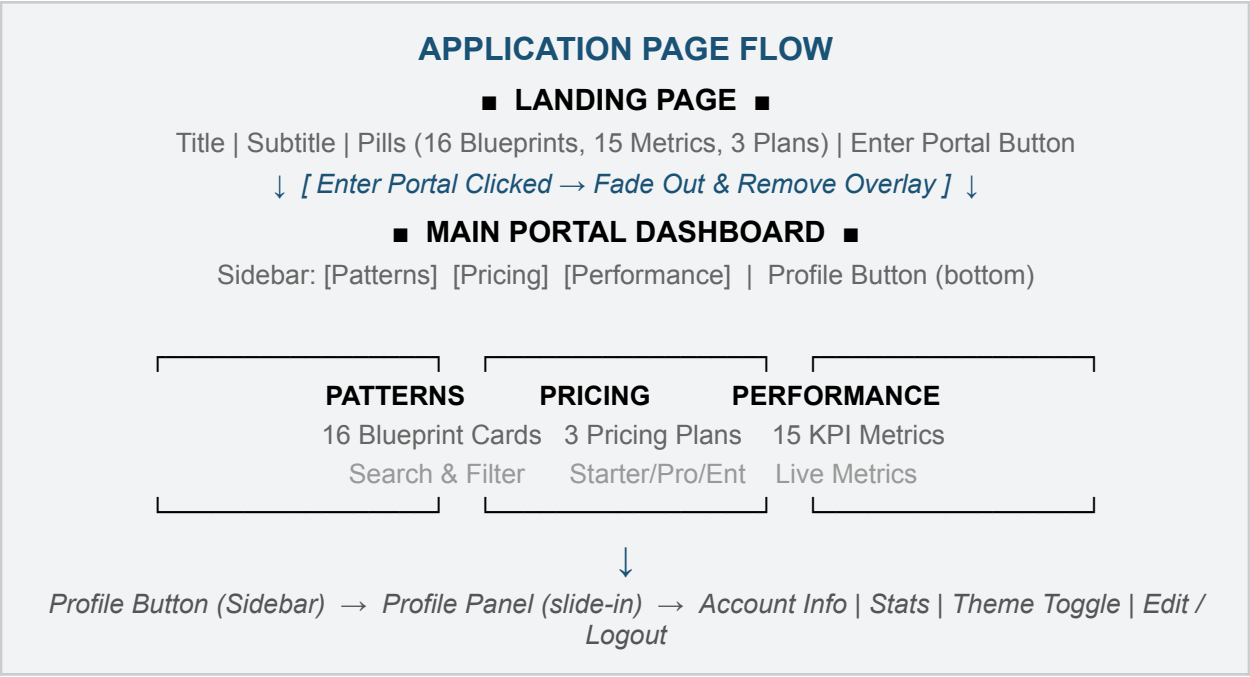
4.3 Visual Design Language

- Typography: Plus Jakarta Sans for all UI text; JetBrains Mono for code-like data (tech stacks, metrics, pricing).
- Dark Mode Palette: Background #080808, Surface #111111, Accent Green #00FFA3, Blue Accent #00D1FF, Yellow #FFCC00.

- Light Mode Palette: Background #F4F4F6, Surface #FFFFFF, consistent accent colours maintained.
- Cards: 24px border radius, subtle glass-morphism borders, hover translateY(-10px) with accent glow.
- Transitions: Smooth cubic-bezier(0.4, 0, 0.2, 1) easing on all interactive elements for a premium feel.

5. Wireframe & Page Flow Diagram

The following diagram describes the logical flow of the DevDocs application from initial load through all interactive states:



6. HTML Structure Explanation

The HTML document (index.html) is semantically organised with clearly defined regions. The structure separates layout, navigation, content, and overlay components:

Element / Section	ID or Class	Purpose
Landing Overlay	#landing	Full-screen intro; fades out and is removed on portal entry
Menu Button	#menuBtn	Fixed floating button to collapse or expand the sidebar

Element / Section	ID or Class	Purpose
Sidebar	#sidebar / .sidebar	Fixed navigation with logo, nav links, and profile button at bottom
Content Wrapper	#content / .content	Hosts the three data sections; shifts with sidebar collapse
Patterns Section	#patterns	Architecture Library; dynamically rendered 16 blueprint cards
Performance Section	#performance	Performance Tuning; dynamically rendered 15 KPI metric cards
Pricing Section	#pricing	Subscription Plans; dynamically rendered 3 pricing tier cards
Profile Panel	#profilePanel	Right-side slide-in panel with user info, stats, and settings
Profile Overlay	#profileOverlay	Backdrop overlay behind the profile panel for focus effect

7. CSS Styling Explanation

7.1 CSS Custom Properties

All theme colours are defined as CSS custom properties on :root, enabling instant full-portal theme switching by changing a single attribute on the <html> element. Key variables:

- --bg, --surface, --card-bg: Layered background system for depth
- --accent (#00FFA3): Primary brand green for CTAs, highlights, and active states
- --accent-blue (#00D1FF): Secondary accent for tech stack labels
- --accent-perf (#FFCC00): Yellow for the Performance section
- --text-main, --text-dim: Two-tier text hierarchy
- --border, --glass: Consistent borders and glass-morphism effects

7.2 Dark / Light Theme Switching

Light mode overrides are scoped to [data-theme="light"], which reassigns surface, border, background, and text variables while preserving accent colours. The .theme-slider element uses CSS transform translateX to animate between states visually. The active theme label is bolded with color: #000 for clear indication.

7.3 Key Layout & Animation Techniques

- Sidebar Collapse: position: fixed; transform: translateX(-100%) with 0.5s cubic-bezier transition.

- Profile Panel: position: fixed; right: -420px transitions to right: 0 when .open class is toggled.
- Cards Grid: CSS Grid with repeat(auto-fill, minmax(350px, 1fr)) for fluid responsiveness.
- Section Transitions: [data-section].active uses a fadeIn keyframe animation (opacity 0→1, translateY 10px→0).
- Hover Effects: Cards lift with transform: translateY(-10px) and display accent-coloured border glow.

8. JavaScript Functionality

8.1 Data Architecture

All content is stored in three plain JavaScript arrays at the top of script.js: patternsData (16 objects), perfData (15 objects), and pricingData (3 objects). This clean data-layer separation means UI rendering functions receive data as parameters, making the code modular and easy to extend.

8.2 Dynamic Rendering Functions

- drawPatterns(list): Maps the patternsData array to HTML template literals and injects the result into #patternsGrid. Accepts a filtered subset for the search feature.
- drawPerf(list): Maps perfData to performance KPI card HTML and injects into #perfGrid.
- drawPricing(): Maps pricingData to pricing card HTML, conditionally applying the .pricing-card--featured class and badge for the Pro plan.

8.3 Interactive Features

Feature	Function(s)	Description
Portal Entry	enterPortal()	Sets landing opacity to 0, removes element after 400ms
Sidebar Toggle	toggleSidebar()	Toggles .collapsed on #sidebar and .full-width on .content
Section Navigation	nav(id, el)	Removes .active from all sections, adds to target; updates nav link
Profile Open	openProfile()	Adds .open to panel and overlay; sets body overflow: hidden
Profile Close	closeProfile()	Removes .open; restores body scroll
Theme Toggle	toggleTheme()	Flips data-theme attribute on <html>; persists choice to localStorage
Real-time Search	input event on #patternSearch	Filters patternsData by name, scale, tech, or latency on each keystroke

Feature	Function(s)	Description
Empty State UI	Conditional in search handler	Renders a branded no-results message when filtered array is empty
Theme Restoration	IIFE on page load	Reads localStorage key 'devdocs-theme' to restore saved preference
Initial Render	window.onload	Calls drawPatterns, drawPerf, and drawPricing to populate all sections

9. Architecture Blueprints — Full Data Overview

The following table summarises all 16 architecture patterns available in the portal, progressing from small-scale classical patterns to enterprise-grade distributed systems:

#	Pattern Name	Tag	Scale	Latency	Key Technology	Cost / Month
1	Monolith	Classic	1K users	Low	Node.js, PostgreSQL	₹1,260
2	MVC Pattern	UI Arch	5K users	Low	Ruby on Rails	₹2,100
3	Layered Architecture	Enterprise	10K users	Medium	Java, Spring Boot	₹3,360
4	Hexagonal	Clean Code	50K users	Low	Python, FastAPI	₹5,040
5	Micro-kernel	Modular	80K users	Low	C++, Qt	₹10,080
6	Serverless FaaS	Cloud	200K users	Variable	AWS Lambda	₹21,000
7	CQRS	Data Logic	400K users	Low	Kafka, .NET	₹37,800
8	Event Sourcing	Immutable	600K users	Medium	EventStoreDB	₹46,200
9	Pub-Sub	Async	800K users	Low	RabbitMQ, Redis	₹58,800
10	Microservices	Distributed	1M users	Medium	Go, Kubernetes	₹1.01L
11	API Gateway	Edge	5M users	Low	Kong, Nginx	₹1.68L
12	Saga Pattern	Transactions	10M users	High	Java, Temporal	₹2.94L

#	Pattern Name	Tag	Scale	Latency	Key Technology	Cost / Month
13	DB Sharding	Database	25M users	Medium	Vitess, MySQL	₹4.20L
14	Cell-Based Architecture	Resilient	50M users	Medium	AWS, Terraform	₹10.08L
15	Sidecar Pattern	Mesh	50M+ users	Low	Istio, Envoy	₹12.60L
16	Bulkhead	Stability	100M users	High	Hystrix, Resilience4j	₹15.12L

10. Pricing Strategy

DevDocs implements a three-tier subscription model designed to capture users at different stages of their professional journey — from students to enterprise teams:

Plan	Price	Target Audience	Key Features
Starter	₹749 / month	Solo engineers, students	5 Blueprints, Community Discord, Monthly Digest, Basic Metrics, 1 Saved Blueprint
Pro	₹2,449 / month	Working professionals	All 16 Blueprints, Trade-off PDF, Code Case Studies, Real-time Dashboard, Unlimited Saves, Priority Support
Enterprise	₹8,299 / month	Teams, organisations	Everything in Pro + 20 Team Seats, Private Slack, 1:1 Architecture Review, Custom Blueprints, Account Manager

11. Performance Metrics Dashboard

The Performance Tuning section presents 15 real-time system KPI cards. Each card shows a metric name, category tag, current value, description, and status indicator. The section uses yellow (#FFCC00) as its accent colour to visually differentiate it from the Patterns section.

Metric Name	Tag	Value	Status
Latency P99	Speed	42ms	Optimized
Throughput	Traffic	120k rps	Stable
Cache Hit Ratio	Memory	94.2%	High
DB Query Time	Database	12ms	Healthy
Error Rate	Stability	0.01%	Excellent

Metric Name	Tag	Value	Status
CPU Utilization	Compute	65%	Normal
Cold Start	Lambda	180ms	Warning
Network Ingress	Bandwidth	4.2 GB/s	Peak
Memory Leak	Audit	Clean	Passed
Disk I/O	Storage	15k IOPS	Optimal
SSL Handshake	Security	8ms	Fast
Message Lag	Kafka	25ms	Low
Hydration Time	Frontend	1.2s	Slow
Goroutine Count	Concurrency	5k active	Stable
Container Boot	Runtime	2.5s	Healthy

12. Key Features Summary

Feature	Implementation Detail
Landing Page	Full-screen animated overlay with gradient title, pill badges, and fade-out entry
Sidebar Navigation	Fixed sidebar with animated collapse; profile button pinned at bottom
Architecture Library	16 dynamically generated blueprint cards with full technical detail
Real-time Search	Instant filtering by name, scale (e.g. 1M users), latency, and tech stack
Empty State UI	Branded no-results message with search icon when no patterns match
Performance Dashboard	15 KPI metric cards with colour-coded status and refresh button
Pricing Plans	3-tier pricing with featured card glow effect and animated CTA buttons
Profile Panel	Slide-in overlay with user stats, account info, and theme toggle
Dark / Light Mode	Full CSS variable theme switching with localStorage persistence
Responsive Layout	Sidebar collapses and pricing grid stacks on mobile viewports

13. Program Output Screenshots

The following screenshots capture the key functionalities of the DevDocs portal:

- Screenshot 1 — Landing Page: Full-screen view with title gradient, subtitle, pill badges (16 Blueprints, 15 Metrics, 3 Plans), and Enter Portal CTA button.

- Screenshot 2 — Architecture Library (Dark Mode): 16 blueprint cards in a responsive grid; each card shows tag, title, description, tech stack, trade-offs, scale, latency, and cost.
- Screenshot 3 — Search Filter in Action: Real-time filtering demonstrated; typing '1M' in the search bar narrows results to the Microservices card.
- Screenshot 4 — Empty State: Branded no-results message with emoji and guidance text when search yields no matches.
- Screenshot 5 — Performance Tuning Section: 15 KPI cards with yellow accent, status badges, and Refresh Metrics button.
- Screenshot 6 — Pricing Plans: Three-column pricing grid; Pro plan highlighted with blue glow and 'Most Popular' badge.
- Screenshot 7 — Profile Panel: Slide-in panel showing user name, Pro Member badge, account info rows, activity stats (42 blueprints read, 8 saved, 3 reviews), and dark/light theme toggle.
- Screenshot 8 — Light Mode: Full portal rendered in light mode; white surfaces with consistent accent colours.

14. Conclusion

DevDocs successfully demonstrates the design and development of a professional, production-quality web application using only native HTML5, CSS3, and Vanilla JavaScript. The project fulfils all requirements of Case Study No. 50 by implementing a complete, functional subscription portal for software architects with the following achievements:

- Semantically structured HTML with clear separation of landing, navigation, content, and overlay regions.
- A sophisticated CSS architecture using custom properties for full, instant theme customisation.
- A data-driven JavaScript layer that dynamically renders all 16 + 15 + 3 cards from structured arrays.
- Responsive design working correctly across desktop and mobile viewports.
- Rich interactivity: real-time search filtering, animated slide-in panels, theme persistence, and smooth CSS transitions.
- A realistic pricing model and live performance dashboard that simulate a genuine SaaS product experience.

The application demonstrates strong understanding of client-side web development principles including DOM manipulation, event handling, CSS animations, responsive layout, and state persistence — all achieved without any external framework or library dependency.